Studio Session 2026-11-01

OpenFlight on the IWR6843 self-trigger, hitting into a net beside a TrackMan 4. 127 IWR triggers over four runs, matched shot by shot against 76 TrackMan shots exported with SwingSync.

TL;DR

- **The OPS243 read 1.49 × TrackMan's ball speed from 11:37:53 until a restart at 12:35.** Its capture window grew from 34 to 48 ms at the same moment: it looks like the radar dropped from 30 to ~20 kS/s mid-run. Outside that period, ball speed matched TrackMan to 0.995 ± 0.028.
- **The self-trigger caught 57 of 76 real shots on time (75 %).** Six fired on the takeaway or backswing, ten never fired, three fired 60-70 ms too early.
- **False triggers are frequent and SNR does not fix them.** Addresses, waggles, practice swings, placing a ball and walking up all fired it at SNR 1.5, 2.0 and 3.0; each one blinds the IWR for 7.7 s and can cost the next real shot.
- **IWR launch was 2.4° MAE (n=8)**, three times the project's 0.83° TrackMan baseline. `--iwr6843-net-m` must be unit → net; entering ball → net (3.0 m) clipped every track.
- **Real-time scheduling buys nothing:** S! reaches the OPS in 0.17 ± 0.01 ms either way.

57 / 76
real shots captured on time

×1.49
OPS ball speed, 11:37:53-12:34 (27 shots)

61
false shots and no-ball triggers with no TrackMan shot

2.4°
IWR launch MAE, against a 0.83° baseline

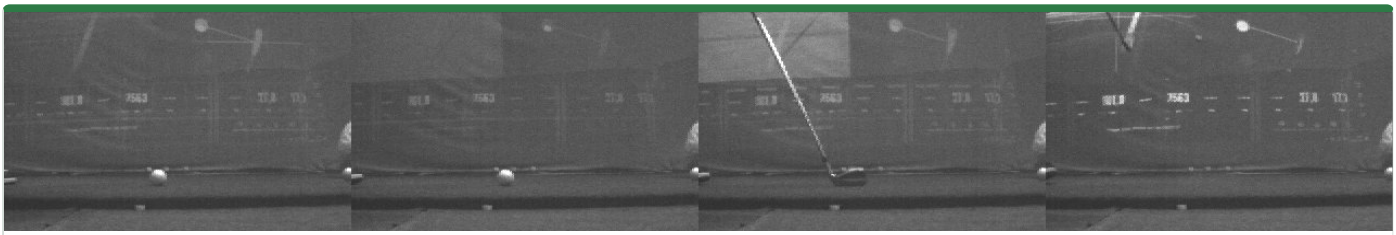
0.17 ms
S! → OPS, normal or real-time

Lessons learned

1. **The OPS changed its sampling mid-run.** At 11:37:53, with no host command in between, ball speed went to 1.486 ± 0.035 × TrackMan, club speed to 1.37 ×, the J3 post-trigger window from ~32 to ~48 ms, the log's `post-trigger` from 34.5 to 0.0 ms, and the clock sync from 36 to 59 valid readings. A restart fixed it.
2. **`--iwr6843-net-m` is unit → net.** 3.0 m (meant as ball → net) left the tracker 0.5 m, about 10 ms of flight. Replaying the same dumps with 4.6 m tracked 24-41 ms on every real shot. Unit → net was 5.30 m.
3. **The self-trigger caught 75 % of real shots on time.** Early triggers on the takeaway left the OPS reading the slow backswing (20-50 mph) and the real shot fell in the blind window.
4. **False triggers are frequent; SNR does not fix them.** 20 saved shots and 41 no-ball triggers had no TrackMan shot behind them. At SNR 3.0 a practice swing still triggered and real shots went missing.
5. **Every trigger blinds the IWR for ~7.7 s** with `--debug`: 7.1 s readback plus a radar restart.
6. **IWR launch angles were 3× worse than the baseline**: 2.4° MAE, 1.4° median, +1.5° bias on 8 clean shots, with the radar 0.04 m above the floor, lower than any validated mount.
7. **Real-time scheduling buys nothing.** The window jitter that looked halved was the 11:37:53 switch.
8. **“TI radar capture failed” is misleading.** It shows on every shot because the firmware never has an onboard result ready; the data is still saved.
9. **Launch OpenFlight from the desktop icon.** Over SSH with `nohup … &` the server ignored SIGINT, and SIGTERM skipped the OPS idle step.

What the camera saw

Each strip shows four frames: 89 ms and 22 ms before the trigger, the trigger frame, and 44 ms after. OV9281 at 320×200, 150 ms before and 50 ms after each IWR trigger.



CAPTURED

Real shot, captured on time

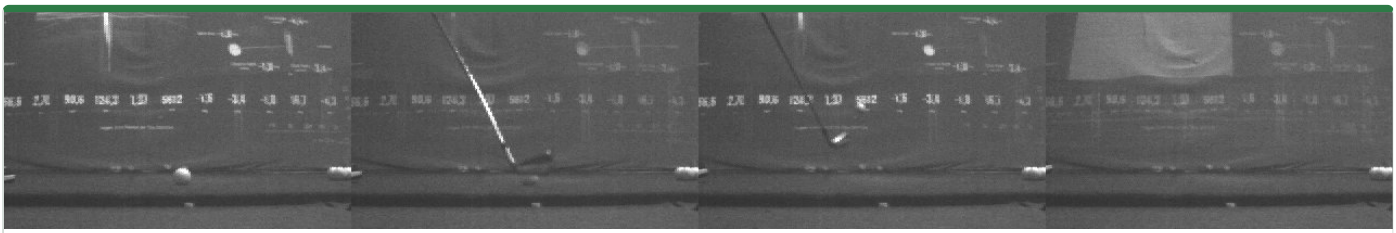
11:32:50 · OpenFlight 120.5 mph, TrackMan 116.4 mph · IWR launch 23.3°, TrackMan 21.3°



WRONG SPEED

Real shot, OPS reading ×1.49

12:03:20 · OpenFlight 192.0 mph, TrackMan 123.3 mph · inside the 11:37:53-12:34 period



CAPTURED

Real shot after the restart

12:49:39 · SNR 3.0 · OpenFlight 122.2 mph, TrackMan 118.8 mph



LOST OR FALSE

Trigger on the takeaway

12:00:52 · fired 0.79 s before TrackMan's impact · saved as 20.4 mph, the real 114.8 mph shot was lost



LOST OR FALSE

Real shot, trigger 70 ms early

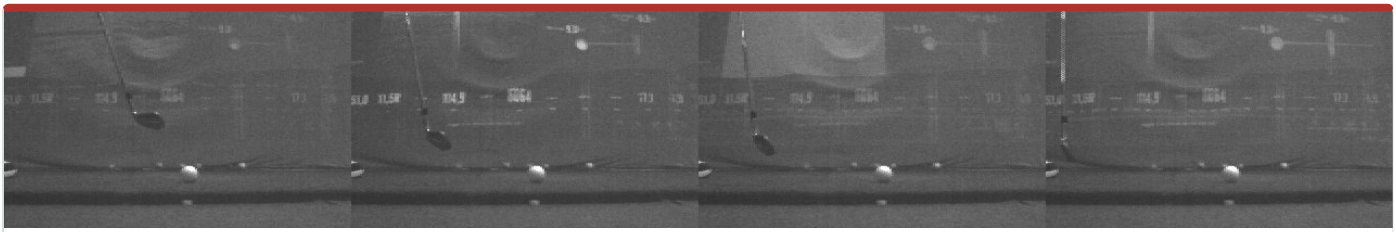
12:48:19 · both the OPS and the camera window closed before impact · TrackMan 121.9 mph, OpenFlight nothing



LOST OR FALSE

Address

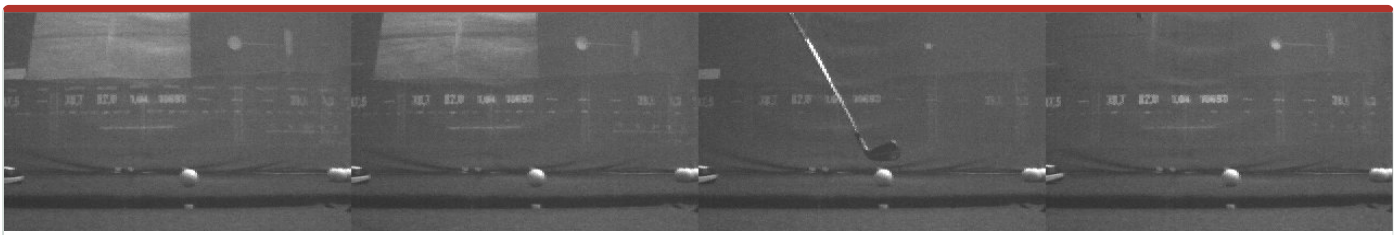
12:15:10 · club resting behind the ball · saved as false shot #36, 20.3 mph from 3 readings



LOST OR FALSE

Practice swing

12:02:40 · ball never moves · saved as false shot #26, 22.5 mph from 1 reading



LOST OR FALSE

Practice swing at SNR 3.0

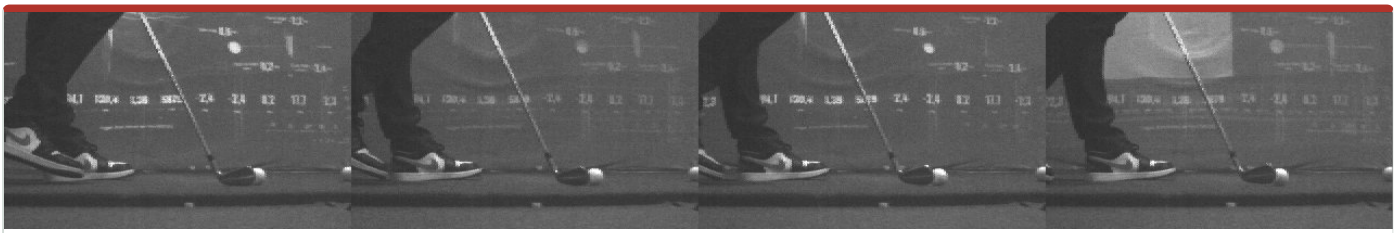
12:48:01 · saved as false shot #2, a 74.0 mph “ball” from 29 readings



LOST OR FALSE

Walking up to the screen

12:04:43 · saved as false shot #30, 27.0 mph from 4 readings



LOST OR FALSE

Placing a ball

12:53:22 · OPS saw no ball, so no shot, but the IWR was blind for 7.7 s

Against TrackMan

TrackMan's clock ran 425.5 s ahead of the Pi's NTP time; at that offset 66 of its 76 shots line up with an IWR trigger within 1 s. TrackMan stamps a shot 70 ± 29 ms after OpenFlight's trigger, so a trigger more than 0.3 s before TrackMan's stamp is an early one.

Did each real shot get captured?

PERIOD	TRACKMAN	ON TIME	EARLY	NO TRIGGER	ON TIME, NO BALL
SNR 1.5, net 3.0 · 11:15-11:31	8	7	0	0	1
SNR 1.5 · 11:32-11:37	6	5	0	1	0
SNR 1.5, OPS ×1.49 · 11:37-12:34	44	31	6	6	1
SNR 2.0 · 12:35-12:45	5	4	0	1	0
SNR 3.0 · 12:45-12:55	13	10	0	2	1
Total	**76**	**57**	**6**	**10**	**3**

Six on-time captures still got a wrong OPS speed (0.14-0.71 × TrackMan). OpenFlight shots with no TrackMan shot behind them: 11 at SNR 1.5, 7 at SNR 2.0, 2 at SNR 3.0.

Accuracy against the project's validation

MEASURE	THIS SESSION	PROJECT BASELINE
OPS ball speed − TrackMan, before cosine correction	−0.5 mph bias, 2.6 mph MAE (n=24)	−2.15 mph bias, 2.83 mph MAE (128 shots)
OPS club speed − TrackMan	−0.4 mph bias, 4.9 mph MAE, p50 2.7 (n=24)	—
IWR LCMF launch − TrackMan, irons	+1.5° bias, 2.4° MAE, p50 1.4° (n=8)	−0.04° bias, 0.83° MAE, p50 0.67° (87 shots)
LCMF coverage on captured shots	24 of 57 (42 %)	87 % strict, irons and wedges

Baseline from OpenFlight's launch-angle report and `speed_correction.py`. LCMF takes OPS speed as an input, so the 16 results inside the ×1.49 period or the net-3.0 run (2.6° MAE) are not a fair test.

The ×1.49 period

	BEFORE 11:37:53, AFTER 12:35	11:37:53-12:34
OPS / TrackMan ball speed	0.995 ± 0.028 (n=24)	1.486 ± 0.035 (n=27)
OPS / TrackMan club speed	1.00 ± 0.11	1.37 ± 0.13
J3 window (S#24 post-trigger)	30-34 ms	46-51 ms
Log `post-trigger`	34.5 ms	0.0 ms

It started between two shots with nothing logged but a normal rearm, and survived the move next to the TrackMan at ~11:52, so it is not TrackMan proximity.

Runs

START	TIME	SNR	NET	TRIGGERS	SHOTS	NO-BALL
111546	11:15-11:31	1.5	3.0 m ✗	20	10	10
113221	11:32-12:34	1.5	5.30 m	76	49	27
123514	12:35-12:45	2.0	5.30 m	17	11	6
124549	12:45-12:55	3.0	5.30 m	14	12	2

“No-ball” is the UI's “impact detected” without metrics. Real-time scheduling was on from 12:13:44 to 12:34.

Net distance

Offline replay of the IWR ball tracker on the dumps from 11:18-11:27, which ran with `--iwr6843-net-m 3.0`:

DUMP	NET 3.0	NET 4.6	NO NET
`111824_001`	reject, 9 ms	ok, 27 ms	ok, 27 ms
`111911_003`	reject, 9 ms	ok, 41 ms	ok, 43 ms
`112540_015`	reject, 13 ms	ok, 41 ms	ok, 41 ms
`112630_017`	ok, 16 ms	ok, 27 ms	ok, 27 ms

Trigger latency and the real-time A/B

OPS J3 pin 1 was wired to the Pi's GPIO18 and logged with `gpiomon`; each camera capture records the host's trigger time on the same monotonic clock.

WINDOW	SCHEDULING	TRIGGERS	S! → OPS	OPS WINDOW
11:32-12:13	normal	54	0.17 ± 0.01 ms	44.8 ± 6.1 ms (30-51)
12:13-12:35	`chrt -a -f 50`	22	0.17 ± 0.01 ms	47.5 ± 2.5 ms (41-51)
12:35-12:45	normal, new process	17	0.16 ± 0.01 ms	31.7 ± 1.8 ms (27-34)
12:45-12:55	normal, new process	14	0.17 ± 0.01 ms	32.1 ± 1.1 ms (31-34)

Impact → S! from the server log: 0.3 ± 3.6 ms (n=44) normal, −0.4 ± 2.9 ms (n=12) real-time, dominated by the impact estimate. Verdict: do not make real-time priority a default.

Setup

OpenFlight	`feat/iwr-self-trigger` + 2 local commits (OPS idle, backlight)
IWR6843 firmware	1.0.3 (git 4e4e9c5d94b7, hybrid-cadence, built 2026-10-09)
OPS243-A	firmware 1.3.3, GPIO UART at 230400 baud, rolling buffer S#24, 30 kS/s
Trigger	IWR self-trigger `triggerCfg 36 <snr> 1`, default offset 0.2 m, no sound sensor
Geometry	ball 1.57 m, net 5.30 m, IWR board centre 0.04 m above the floor, tilt 10.1°
Camera	OV9281, 320×200 at 450 fps (430.9 delivered, 0 gaps on 127 captures)
Reference	TrackMan 4, exported with SwingSync
Storage	~6.6 MB per trigger with camera and `--debug`; 0.83 GB for this session





Next studio session

- **Before the kiosk:** sweep the self-trigger with `scripts/iwr6843/swing_trigger.py` at SNR 1.5, 2.0 and 3.0. It measures the lane's noise floor, arms the trigger and prints PASS or FAIL per swing, with no OPS and no 7.7 s blind window. Count fires for real swings, addresses, waggles, takeaways and walking past.
- **Robinante's log tails** in two extra SSH terminals for the whole session (forgotten this time).
- **Self-trigger flags**, one block each: `--iwr6843-self-trigger-offset-m` 0.4 and 0.6, `--iwr6843-ball-detector` on and follow (drives the “ball placed” banner), `--iwr6843-veto-no-ball`, then `self-trigger-bin`, SNR and `ball-snr`.
- **Watch the J3 window live** for the ×1.49 switch (34 → 48 ms) and query the OPS before restarting when it happens.
- **Robinante's offset test** against TrackMan interference: `?F`, `T=-2`, wait 15 s, `?F`, with OpenFlight stopped.
- **A block without `--debug`** to shorten the blind window, and SwingSync export again for `python -m openflight_bench.session compare`.

Data, tools and the full report: HuggeK/OpenFlight-Radar-Workbench, `docs/reports/studio-session-2026-11-01`. Numbers come from the kiosk logs, the J3 edge log, the camera metadata and the SwingSync export, through `python -m openflight_bench.session`.